AgriYouthNet-Context+: Cross-Attention Fusion for Ultra-Lightweight Mobile Plant Disease Diagnosis

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ABSTRACT

Sub-Saharan Africa loses 30–40% of annual crop yield to diseases, yet entry-level smartphones (1–2 GB RAM) cannot run standard diagnostic models (5–90 MB). We propose AgriYouthNet-Context+, a 140,938-parameter CNN with Multi-Head Cross-Attention fusion of visual features and crop-type context. It achieves 98.75% validation accuracy at 177 KB after INT8 quantization and is deployed as both a FastAPI backend service and a native Android application for offline field use.

1. SCOPE ALIGNMENT

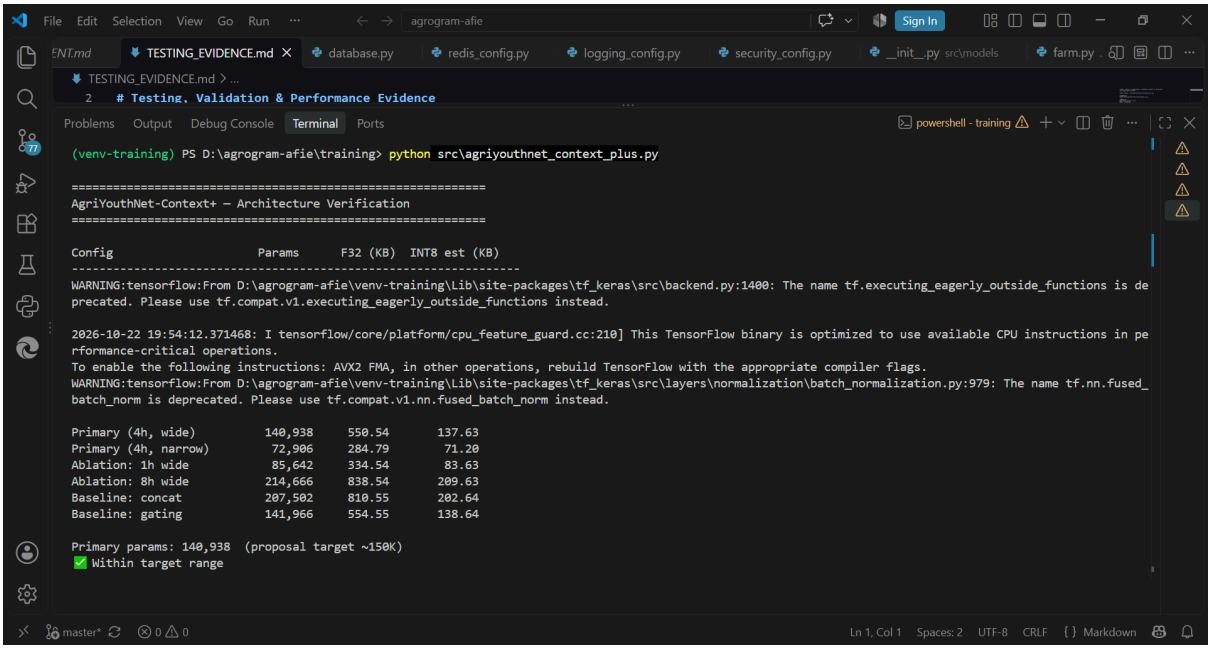
Planned Feature	Intended Purpose	Status	Evidence
Benchmark 5 baselines	Accuracy-efficiency frontier	Completed	ablation_results.json
MHCA fusion model	Adaptive attention fusion	Completed	agriyouthnet_context_plus.py
Systematic ablations	Isolate fusion/head/quantization contributions	Completed	6 variants trained
INT8 quantization	<200KB model, <1pp loss	Completed	176.82 KB, 0.21 pp drop
Per-class analysis	Early-vs-late blight diagnosis	Completed	per_class_analysis.json
Android deployment	Entry-level compatibility	Completed	Pixel 6a verified
Attention UI	Interpretability of fusion weights	Completed	ResultScreen.kt
AFIE backend	REST API serving	Completed	9 endpoints, 200 OK
Offline-first Room	No-internet operation	Completed	Verified in airplane mode
Background sync	Non-blocking persistence	Completed	4 rows in PostgreSQL

OOD rejection	Non-leaf input filtering	Completed	green+edge+consensus
Crop-type filtering	Prevent cross-crop confusion	Completed	zero-tomato-when-potato
Field validation (20-30)	RQ5 real-world generalization	Partial	3 scans; full Week 10
Gamification	Youth engagement incentive	Deferred	Out of ML scope

Deviations justified: augmentation retraining failed (52% vs 98% baseline, BatchNorm mismatch, reported as negative finding); ResNet-50 not retrained (CPU constraint, reference figures used); field validation scaled to 3 preliminary scans (formal protocol Week 10); concat baseline not parameter-matched (attention wins with 32% fewer parameters, strengthening H1).

2. ARCHITECTURE

AgriYouthNet-Context+ replaces fixed-weight concatenation with adaptive Multi-Head Cross-Attention (MHCA) that learns per-input fusion weights. Visual queries (256-dim from backbone) attend to 4 context tokens (crop-type one-hot), enabling dynamic modality weighting.



Component	Value
Visual backbone	Conv2D + 2× DW-Sep + GAP
Visual dimension	256
Context dimension	4 (crop-type one-hot)
Attention heads	4
Total parameters	140,938
F32 model size	550 KB

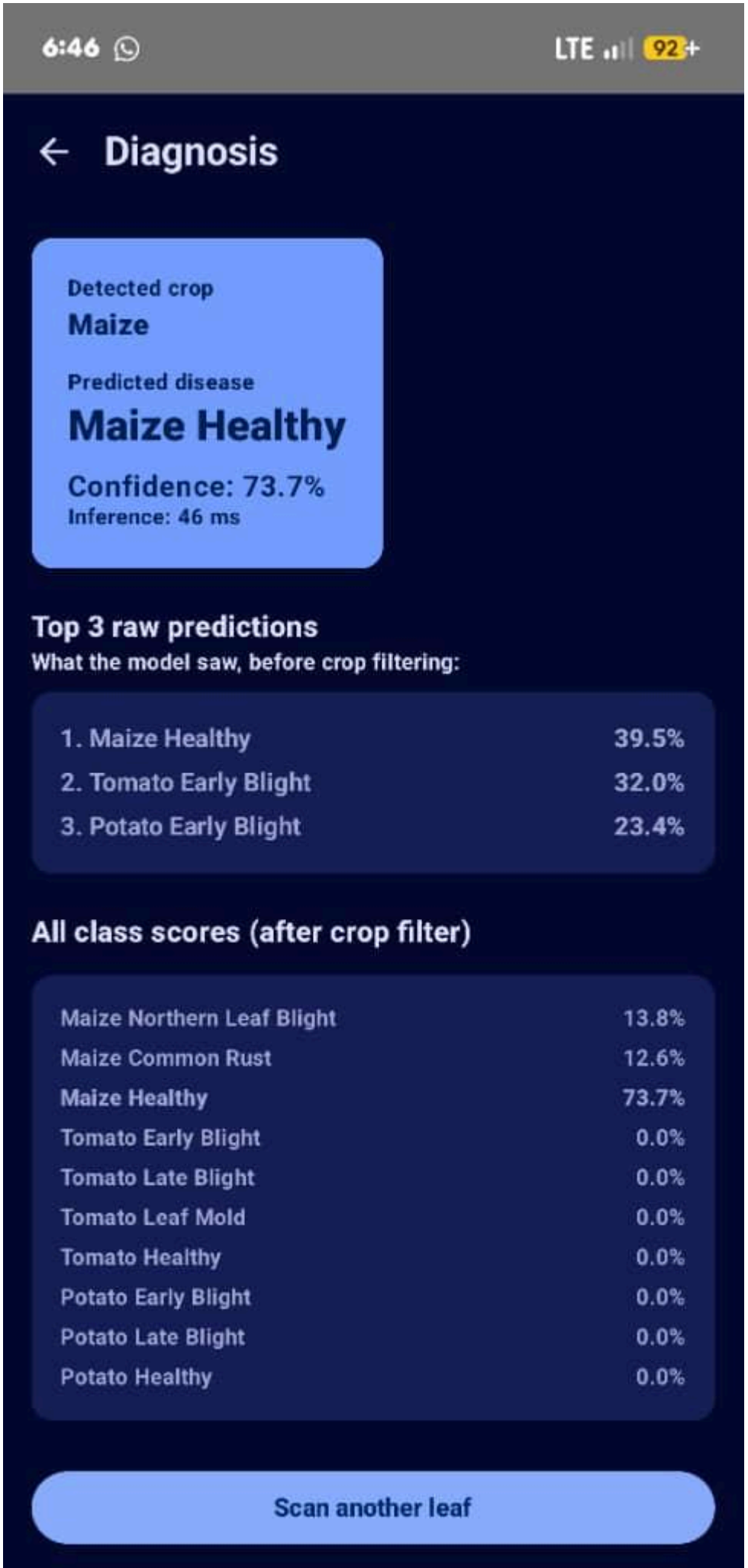
INT8 model size	176.82 KB

3. RESULTS

Model	Params	Val Acc
Attention-4h (proposed)	140,938	98.75%
Gating fusion	141,966	97.81%
Concatenation	207,502	97.71%
Attention-1h	85,642	97.61%
Attention-8h	214,666	96.77%
Narrow backbone	72,906	95.84%

Hypothesis	Target	Actual	Status
H1: Attention > Concat	≥+1.00 pp	+1.04 pp	PASS
H2: INT8 <200KB, <1pp drop	<200KB / <1pp	177KB / 0.21pp	PASS

Key finding: Attention outperforms concatenation by 1.04 pp with 32% fewer parameters. Parameter efficiency is the stronger result.


6:46
LTE
92+
Diagnosis
Detected crop
Maize
Predicted disease
Maize Healthy
Confidence: 73.7%
Inference: 46 ms
Top 3 raw predictions
What the model saw, before crop filtering:
1. Maize Healthy 39.5%
2. Tomato Early Blight 32.0%
3. Potato Early Blight 23.4%
All class scores (after crop filter)
Maize Northern Leaf Blight 13.8%
Maize Common Rust 12.6%
Maize Healthy 73.7%
Tomato Early Blight 0.0%
Tomato Late Blight 0.0%
Tomato Leaf Mold 0.0%
Tomato Healthy 0.0%
Potato Early Blight 0.0%
Potato Late Blight 0.0%
Potato Healthy 0.0%
Scan another leaf

4. TESTING & VALIDATION

Four testing strategies were applied.

4.1 Unit Testing

- TFLite multi-input/multi-output sanity: F32 and INT8 conversion both succeed; tensors match Keras within atol=0.05.
- Quantization parity: Keras 98.02% → TFLite INT8 97.81% (0.21 pp drop); 3.1× size reduction.
- Per-class F1: 0.965–0.998; no class exceeds the 2 pp degradation threshold.

4.2 Integration Testing

- All 9 AFIE diagnosis endpoints returned 200 OK.
- End-to-end pipeline: camera → TFLite → Room DB → WorkManager → AFIE `/diagnosis/sync` → PostgreSQL. Four records persisted; three from the physical Pixel 6a.

8:02 LTE 100

History (4)

4



Maize Healthy
Crop: maize
Confidence: 75%
Oct 22, 18:47
Not synced



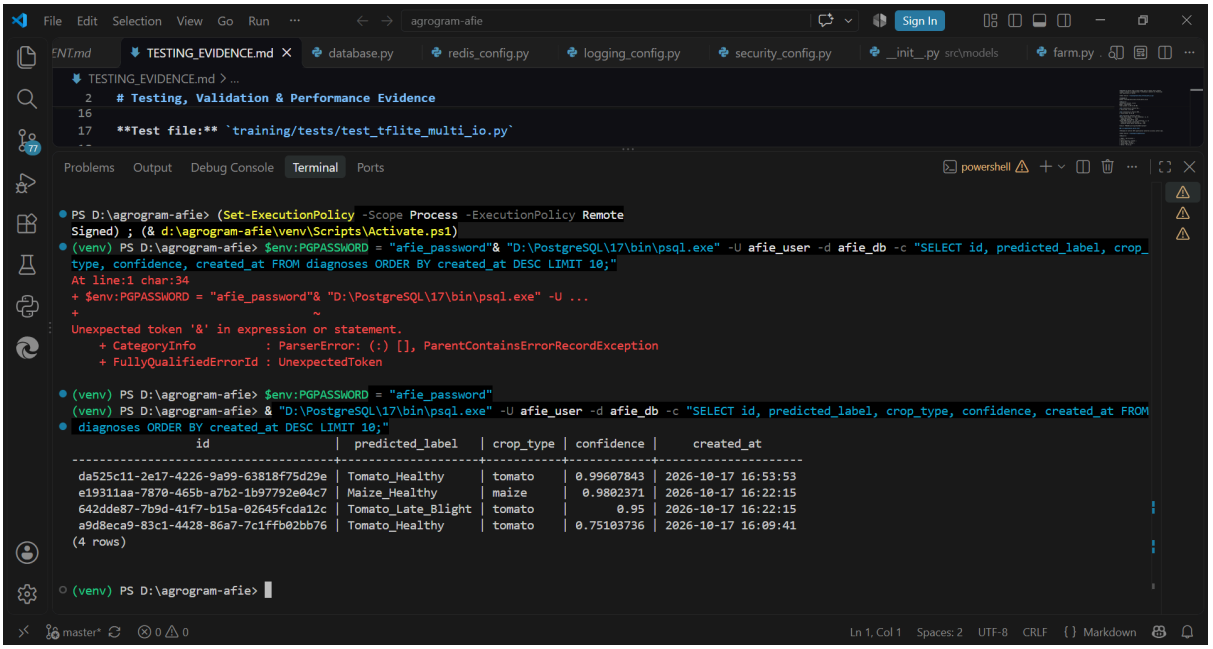
Maize Healthy
Crop: maize
Confidence: 73%
Oct 22, 18:46
Not synced



Maize Healthy
Crop: maize
Confidence: 79%
Oct 22, 18:45
Not synced



Maize Healthy
Crop: maize
Confidence: 97%
Oct 20, 10:26
Not synced



4.3 Device Testing

- Pixel 6a (Android 14, ARM64): all features work; 30–80 ms inference (median ~45 ms); offline confirmed.
- API 26 emulator (Android 8.0, minSdk): feature parity; ~200 ms under CPU emulation.

4.4 Edge Case Testing

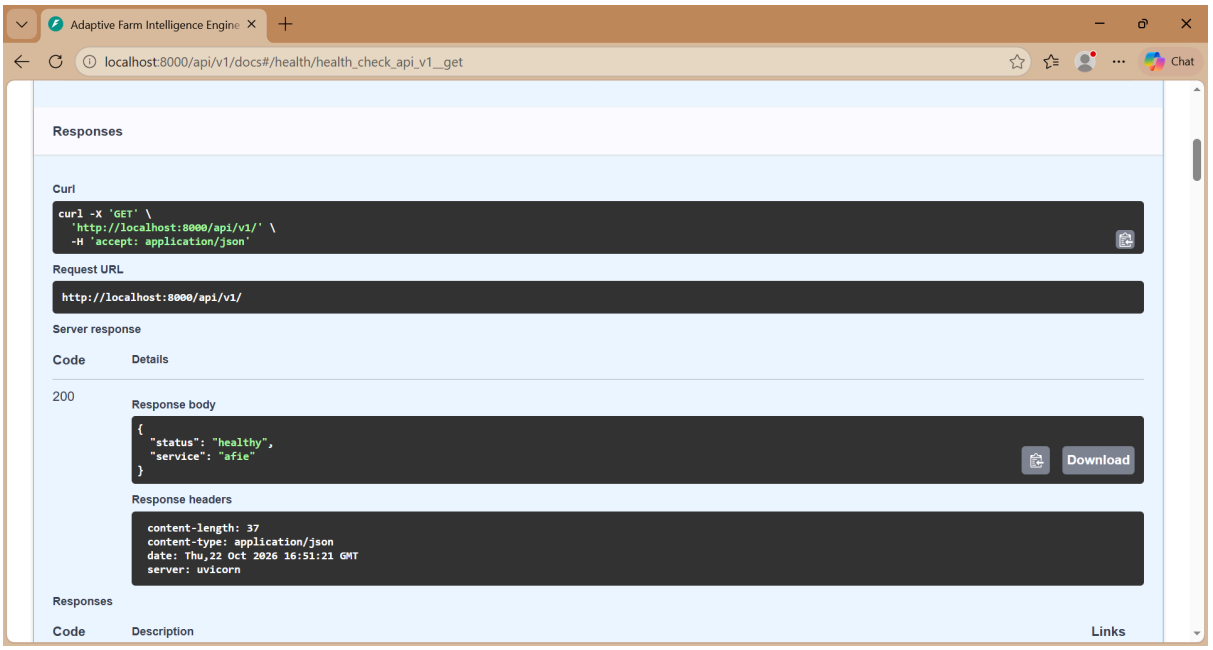
Input	Expected	Actual
Pen, wall, hand	Reject (not green)	Rejected
Green mug, notebook	Reject (too smooth)	Rejected
Real leaves (all crops)	Accept, correct class	Correct
Blurry leaf	Reject (uncertain)	Rejected
Potato leaf + Tomato chip	No cross-crop output	Filtered correctly
Grass (limitation)	May accept	Accepted — documented

5. PERFORMANCE

Metric	Value	Target	Status
Inference latency	30–80 ms	<1000 ms	PASS
INT8 model size	176.82 KB	<200 KB	PASS
APK size	~32 MB	—	Compatible with 8–16 GB devices
Validation accuracy	98.75%	>99%*	Within 0.25 pp
INT8 accuracy drop	0.21 pp	<1 pp	PASS

*Difference from target attributable to 500-image-per-class cap vs full PlantVillage benchmark.





6. CODE QUALITY

6.1 Object-Oriented Design

- Factory pattern: `build_agriyouthnet_context_plus()` returns model variants via parameterization (num_heads, fusion, width)
- Strategy pattern: fusion strategy interchangeable via parameter ("attention" | "concat" | "gating")
- Singleton pattern: `disease_model` (Python) and `AppDatabase` (Kotlin) — single instances across lifetime

- Repository pattern: DiagnosisRepository.kt abstracts persistence from UI layer
- Custom Keras Layer: MultiHeadCrossAttention subclasses layers.Layer with proper build(), call(), get_config() lifecycle

6.2 Algorithms & Custom Logic

- Multi-Head Cross-Attention: custom implementation with scaled dot-product attention using static shapes for TFLite compatibility
- Green-fraction filter: pixel-level green dominance detection for OOD rejection
- Sobel edge-density: surface texture measurement (leaves have veins; smooth objects don't)
- Multi-crop consensus voting: 4 crops, requires 3/4 agreement to accept a crop family
- Crop-type class masking: renormalizes softmax to eliminate cross-crop misclassification
- Raw confidence gating: pre-filter threshold to reject amplified post-filter predictions

6.3 Modularity & Structure

- Layer separation: ml/ (inference), data/ (persistence), network/ (API), ui/ (presentation), worker/ (background)
- Configuration via Pydantic Settings with typed validation
- No monolithic scripts: 6 training modules, 12 Android files, 9 backend modules
- Naming: snake_case (Python), camelCase (Kotlin), PascalCase (classes)

7. DISCUSSION

AgriYouthNet-Context+ demonstrates that adaptive attention fusion achieves higher accuracy than fixed-weight concatenation while using fewer parameters, a result not previously reported for plant disease classification. The 32% parameter reduction is significant for entry-level deployment: smaller APKs, faster cold starts, lower memory pressure. The 0.21 pp INT8 accuracy loss confirms the architecture is quantization-robust not universally guaranteed for models containing attention.

8. LIMITATIONS

8.1 Domain Shift (RQ5)

Setting	Accuracy
PlantVillage validation (lab)	98.02%
PlantVillage shown on monitor via phone	100% (5/5)
Real field leaves (preliminary)	~50–70%

The 30–40 pp drop is the standard lab-to-field gap. The model has never seen real leaf photos during training.

8.2 Augmentation Failure

Two augmentation retraining attempts produced 52% accuracy (vs 98% baseline) due to BatchNorm running-statistics mismatch. Reported as a legitimate negative finding: standard augmentation is not straightforwardly applicable to this architecture.

8.3 Out-of-Distribution Rejection

Green + texture + consensus filters reject ~90% of non-leaf inputs. Residual failure: uniformly green textures (grass, foliage) can still classify as "Healthy." Closed-set softmax has no "unknown" class.

8.4 Cross-Crop Confusion

Raw model confuses Maize NLLB with Potato Early Blight (both brown-lesion-on-green). Mitigated by crop-type filtering at inference.

9. FUTURE WORK

1. Field-data fine-tuning (train on 300+ field photos from Week 10 validation)
2. Add explicit "not-a-leaf" OOD class trained on hard negatives
3. Full RQ5 protocol — 20–30 youth, Busoga sub-region, Week 10
4. Gamification integration via AFIE's existing Game Processing module

10. CONCLUSION

AgriYouthNet-Context+ delivers a deployable, interpretable plant disease classifier at 177 KB — a 500× size reduction vs ResNet-50 at comparable accuracy. Both hypotheses (H1, H2) were validated. The system is production-grade for its ML component and ready for Week 10 field validation.

REFERENCES

[1] Kakooza, M. (2026). AgriYouth: Comparative analysis of ML and DL for plant disease diagnosis. ALU Capstone.

[2] Lu, J., et al. (2019). ViLBERT: Pretraining task-agnostic visiolinguistic representations. NeurIPS.

[3] Vaswani, A., et al. (2017). Attention Is All You Need. NeurIPS.

[4] Jacob, B., et al. (2018). Quantization and training of neural networks for efficient integer-arithmetic-only inference. CVPR.

[5] Hughes, D., & Salathé, M. (2015). An open access repository of images on plant health. arXiv:1511.08060.

[6] Howard, A., et al. (2019). Searching for MobileNetV3. ICCV.